The Lebesgue integral of non-negative functions

REAL VARIABLES I

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For a non-negative function on an interval of $\mathbb{R}$, the Riemann integral measures the area below its graph by summing the areas of thin vertical rectangles. On a general measure space $(X, \mathcal{A}, \mu)$, however, the domain X need not possess a topological structure, an interval structure, or a linear order. Consequently, vertical partitioning is unavailable.

The Lebesgue integral solves this problem by partitioning the subgraph into horizontal slices instead. At height $\lambda \geq 0$, the horizontal slice corresponds to the set of points where $f(x) > \lambda$, and its "area" is measured using μ. In these notes, we construct the Lebesgue integral via these horizontal superlevel sets, establish its agreement with simple-function integrals, and prove its first approximation and linearity properties.

After studying these notes, you will be able to do the following.

- Define the non-negative Lebesgue integral from the measures of its superlevel sets.
- Relate the superlevel-set definition to the finite-sum formula for a simple function.
- Compute the integral of a simple function.
- Define the Archimedean integral for non-increasing functions.
- Prove the monotone convergence theorem for the Lebesgue integral and relate it to the monotone convergence properties of the Archimedean integral.
- Deduce positive linearity from simple-function approximation and monotone convergence.

Throughout these notes, $(X, \mathcal{A}, \mu)$ denotes a measure space and all functions under consideration are measurable. Arithmetic for non-negative integrals takes place in $[0, +\infty]$, with the standard convention $0 \cdot (+\infty) = 0$.

7.1 Definition through superlevel sets

For a non-negative function f and a height $\lambda \geq 0$, the set $\{x \in X : f(x) > \lambda\}$ is called the **superlevel set**. Its measure quantifies the width of the horizontal slice where λ intersects the subgraph of f.

Definition 7.1 (Superlevel distribution function). For a measurable function $f\colon X \to [0, +\infty]$, its **superlevel measure function** (or distribution function) is defined by

$$\lambda \mapsto \mu(\{x \in X : f(x) > \lambda\}), \qquad \lambda \in [0, +\infty).$$

We will often use the shorthand $\mu(\{f > \lambda\})$.

The function $\lambda \mapsto \mu(\{x \in X : f(x) > \lambda\})$ is non-negative and non-increasing on $[0, +\infty)$. Indeed, if $0 \leq \lambda < \eta$, then $\{x \in X : f(x) > \eta\} \subseteq \{x \in X : f(x) > \lambda\}$, so monotonicity of the measure yields $\mu(\{f > \eta\}) \leq \mu(\{f > \lambda\})$.

To define the integral of such a non-increasing function over $[0, +\infty)$, we use lower sums over finite partitions.

Definition 7.2 (Archimedean integral). For a non-increasing function $g\colon [0,+\infty) \to [0,+\infty]$ and a finite partition $P = (0 = \lambda_0 < \lambda_1 < \cdots < \lambda_m)$, the lower sum of g with respect to P is

$$L(g;P) \coloneqq \sum_{i=0}^{m-1} g(\lambda_{i+1})(\lambda_{i+1} - \lambda_i).$$

For each $i = 0, \dots, m-1$, the corresponding term represents the area of a rectangle with width $\lambda_{i+1} - \lambda_i$ and height $g(\lambda_{i+1})$. Because g is non-increasing, $g(\lambda_{i+1}) = \inf_{x \in [\lambda_i, \lambda_{i+1}]} g(x)$, and the rectangle $[\lambda_i, \lambda_{i+1}] \times [0, g(\lambda_{i+1})]$ lies below the graph of g over $[\lambda_i, \lambda_{i+1}]$. The **Archimedean integral** of g is

$$\int_0^{+\infty} g(\lambda)\, \mathrm{d}\lambda \coloneqq \sup_P L(g;P) \in [0,+\infty], \tag{7.1}$$

where the supremum is taken over all finite partitions of $[0, \lambda_m]$ with arbitrary finite endpoint $\lambda_m > 0$.

If $g(t) = +\infty$ for any $t > 0$ it is immediate to check that $\int_0^{+\infty} g(\lambda)\, \mathrm{d}\lambda = +\infty$. On the other hand, if g is finite-valued on $(0,+\infty)$, its monotonicity guarantees that g is Riemann integrable on every compact interval $[a,b] \subset (0,+\infty)$ and satisfies

$$\int_0^{+\infty} g(\lambda)\, \mathrm{d}\lambda = \lim_{a \downarrow 0,\, b \uparrow +\infty} \int_a^b g(\lambda)\, \mathrm{d}\lambda,$$

where the integrals on the right are ordinary Riemann integrals.

We now combine these concepts to define the Lebesgue integral of a non-negative function.

Definition 7.3 (Lebesgue integral of a non-negative function). For a measurable function $f\colon X \to [0,+\infty]$, its **Lebesgue integral** is

$$\int_X f\, \mathrm{d}\mu \coloneqq \int_0^{+\infty} \mu(\{x \in X\colon f(x) > \lambda\})\, \mathrm{d}\lambda, \tag{7.2}$$

where the integral on the right is the Archimedean integral from Definition 7.2.

Formula (7.2) is called the **layer-cake formula**. It expresses the integral of f by accumulating the measures of its superlevel sets.

7.2 Agreement with simple-function integrals

7.2.1 Simple functions

For a measurable set $A \in \mathcal{A}$, its indicator function $\mathbb{1}_A\colon X \to \{0,1\}$ is defined by

$$\mathbb{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

For $a \geq 0$, the function $a\mathbb{1}_A$ takes value a on A and 0 outside A. **Simple functions** are finite linear combinations of indicator functions.

Definition 7.4 (Non-negative simple function). A function $\varphi\colon X \to [0,+\infty)$ is a **non-negative measurable simple function** if it satisfies either of the following equivalent conditions:

1. The range $\varphi(X)$ is finite, and every level set $\varphi^{-1}(\{a\})$ is measurable.

2. There exist measurable sets $E_1, \dots, E_m \in \mathcal{A}$ and numbers $a_1, \dots, a_m \geq 0$ such that $\varphi = \sum_{i=1}^{m} a_i \mathbb{1}_{E_i}$.

Condition (1) lists the distinct values of φ and their corresponding level sets. If $0 < \lambda_1 < \cdots < \lambda_N$ are the positive values of φ, let $A_j = \varphi^{-1}(\{\lambda_j\})$ for $j \geq 1$. Set $\lambda_0 = 0$ and take $A_0 = \varphi^{-1}(\{0\})$ when 0 is a value of φ, or $A_0 = \emptyset$ otherwise. Then

$$\varphi = \sum_{j=0}^{N} \lambda_j \mathbb{1}_{A_j},$$

and the sets $A_0, \dots, A_N$ are measurable, pairwise disjoint, and partition X. This is the canonical representation of φ. Condition (2), in contrast, permits the sets E_i to overlap.

7.2.2 The superlevel formula for simple functions

The following proposition confirms that evaluating the layer-cake formula for a simple function yields the expected sum of values weighted by set measures.

Proposition 7.5 (Layer-cake formula for simple functions). Let $A_0, \dots, A_N$ be a finite *pairwise disjoint* collection of measurable sets. For a non-negative measurable simple function

$$\varphi = \sum_{j=0}^{N} \lambda_j \mathbb{1}_{A_j}, \qquad 0 = \lambda_0 < \lambda_1 < \cdots < \lambda_N,$$

the equality

$$\int_X \varphi \, \mathrm{d}\mu = \sum_{j=0}^{N} \lambda_j \mu(A_j) \tag{7.3}$$

holds.
According to convention, the term corresponding to $\lambda_0 = 0$ is defined to be 0, even when $\mu(A_0) = +\infty$. Recall that $\int_X \varphi \, \mathrm{d}\mu$ is *defined as* $\int_0^{+\infty} \mu(\{x \in X : \varphi(x) > \lambda\}) \, \mathrm{d}\lambda$.

Proof of Proposition 7.5. By Definition 7.3, the Lebesgue integral of φ is given by

$$\int_X \varphi \, \mathrm{d}\mu = \int_0^{+\infty} \mu(\{x \in X : \varphi(x) > \lambda\}) \, \mathrm{d}\lambda.$$

We obtain this Archimedean integral directly from the values of φ.
Expressing each λ_j as a sum of consecutive differences $\lambda_j = \sum_{i=0}^{j-1} (\lambda_{i+1} - \lambda_i)$ and interchanging the finite sums yields

$$\begin{aligned}
\sum_{j=0}^{N} \lambda_j \mu(A_j) &= \sum_{j=1}^{N} \left(\sum_{i=0}^{j-1} (\lambda_{i+1} - \lambda_i) \right) \mu(A_j) \\
&= \sum_{i=0}^{N-1} (\lambda_{i+1} - \lambda_i) \sum_{j=i+1}^{N} \mu(A_j) \\
&= \sum_{i=0}^{N-1} (\lambda_{i+1} - \lambda_i) \mu \left(\bigcup_{j=i+1}^{N} A_j \right).
\end{aligned}$$

For $\lambda_i \leq \lambda < \lambda_{i+1}$, the superlevel set is

$$\{x \in X : \varphi(x) > \lambda\} = \bigcup_{j=i+1}^{N} A_j.$$

Because the distribution function is constant on each interval $[\lambda_i, \lambda_{i+1})$, its Archimedean integral equals the final sum in (7.3). Values at the finitely many boundary points $\lambda_0, \dots, \lambda_N$ do not affect the supremum of lower sums. □

7.2.3 Overlapping indicator representations

We have defined the integral and proved that it has the intuitive meaning we expect. However, we do not yet have one of its most important properties: linearity. We will first establish linearity for simple functions. To do so, we need to know that the values-times-measures sum depends only on the simple function, not on its particular indicator representation. The preceding proposition evaluates this sum for a pairwise-disjoint representation; we now extend it to representations whose indicator sets overlap.

Proposition 7.6 (Well-definedness of the values-times-measures sum). For measurable sets $A_1, \dots, A_m, C_1, \dots, C_n \in \mathcal{A}$ and coefficients $a_1, \dots, a_m, b_1, \dots, b_n \geq 0$, if

$$\varphi = \sum_{i=1}^{m} a_i \mathbb{1}_{A_i} = \sum_{k=1}^{n} b_k \mathbb{1}_{C_k},$$

then

$$\sum_{i=1}^{m} a_i \mu(A_i) = \sum_{k=1}^{n} b_k \mu(C_k). \tag{7.4}$$

Consequently, the common value in (7.4), denoted by $I(\varphi)$, is well defined.

Proof of Proposition 7.6. We partition X according to which sets among $A_1, \dots, A_m$ contain a given point. For each tuple $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m) \in \{-1, 1\}^m$, set $A_i^{(1)} \coloneqq A_i$, $A_i^{(-1)} \coloneqq X \setminus A_i$, and

$$B_\varepsilon \coloneqq \bigcap_{i=1}^{m} A_i^{(\varepsilon_i)}.$$

The sets B_ε are pairwise disjoint and partition X. On B_ε, the function φ takes the constant value $\sum_{\varepsilon_i = 1} a_i$. Grouping cells with identical constant values and applying Proposition 7.5 gives

$$\begin{aligned} \int_X \varphi \, \mathrm{d}\mu &= \sum_{\varepsilon \in \{-1,1\}^m} \left(\sum_{i \colon \varepsilon_i = 1} a_i \right) \mu(B_\varepsilon) \\ &= \sum_{i=1}^{m} a_i \sum_{i \colon \varepsilon_i = 1} \mu(B_\varepsilon) = \sum_{i=1}^{m} a_i \mu(A_i). \end{aligned}$$

The same argument applied to the representation using the sets C_k gives

$$\int_X \varphi \, \mathrm{d}\mu = \sum_{k=1}^{n} b_k \mu(C_k),$$

which proves (7.4). □

Proposition 7.7 (Linearity of the values-times-measures sum). For non-negative measurable simple functions φ, ψ and constants $r, s \geq 0$,

$$I(r\varphi + s\psi) = rI(\varphi) + sI(\psi).$$

Consequently,

$$\int_X (r\varphi + s\psi)\,\mathrm{d}\mu = r\int_X \varphi\,\mathrm{d}\mu + s\int_X \psi\,\mathrm{d}\mu.$$

Proof of Proposition 7.7. Choose indicator representations

$$\varphi = \sum_{i=1}^{m} a_i \mathbb{1}_{A_i} \quad \text{and} \quad \psi = \sum_{k=1}^{n} b_k \mathbb{1}_{C_k}.$$

Then

$$r\varphi + s\psi = \sum_{i=1}^{m} (ra_i)\mathbb{1}_{A_i} + \sum_{k=1}^{n} (sb_k)\mathbb{1}_{C_k}.$$

By the well-definedness just proved,

$$I(r\varphi + s\psi) = r\sum_{i=1}^{m} a_i \mu(A_i) + s\sum_{k=1}^{n} b_k \mu(C_k) = rI(\varphi) + sI(\psi).$$

The canonical representation and Proposition 7.5 show that $I(\eta) = \int_X \eta\,\mathrm{d}\mu$ for every non-negative measurable simple function η, so the stated linearity of the integral follows. □

7.2.4 Example

Let $E = [0,1) \cup [2,4) \cup [4,5)$ equipped with Lebesgue measure, and consider

$$\varphi(x) = \begin{cases} 1, & x \in [0,1), \\ 7, & x \in [2,4), \\ 5, & x \in [4,5). \end{cases}$$

Figure 7.1 displays the subgraph of φ alongside its superlevel distribution function.

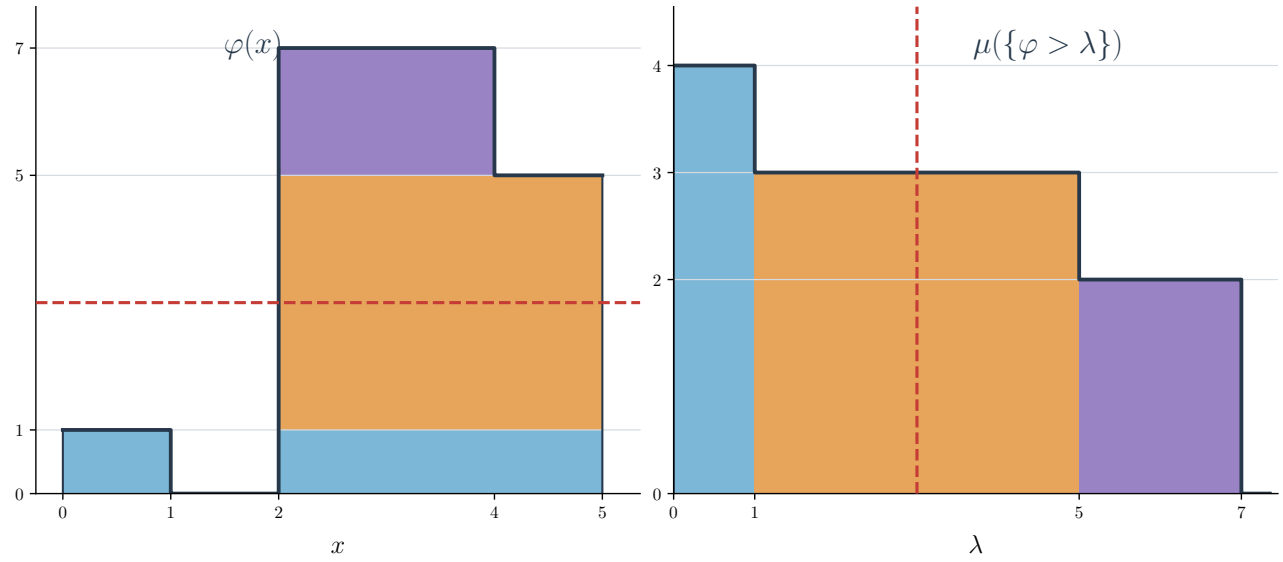


Figure 7.1: The graph of φ (left) and the function $\lambda \mapsto \mu(\{x \in E : \varphi(x) > \lambda\})$ (right). Matching colors represent equal areas in the two panels.

Figure description: Two color-matched plots. The left plot shows a simple function of height 1 on the interval from 0 to 1, height 7 from 2 to 4, and height 5 from 4 to 5. Its subgraph is split into horizontal blue, orange, and purple bands. The right plot shows the distribution function with values 4 from height 0 to 1, 3 from height 1 to 5, and 2 from height 5 to 7; the areas under these steps have the matching colors.

For this function, the superlevel measure function is

$$\mu(\{x \in E : \varphi(x) > \lambda\}) = \begin{cases} 4, & 0 \le \lambda < 1, \\ 3, & 1 \le \lambda < 5, \\ 2, & 5 \le \lambda < 7, \\ 0, & \lambda \ge 7. \end{cases}$$

Evaluating both sides confirms the equality:

$$1 \cdot 1 + 7 \cdot 2 + 5 \cdot 1 = 20 = 1 \cdot 4 + (5 - 1) \cdot 3 + (7 - 5) \cdot 2.$$

7.3 Monotone convergence

The proof of monotone convergence consists of two steps. First, we show that the Archimedean integral respects increasing limits of non-increasing functions. Second, we verify that the measures of horizontal slices of f_n increase pointwise to the measures of horizontal slices of f.

Proposition 7.8 (Monotone convergence for Archimedean integrals). For non-increasing functions $g_n, g \colon [0, +\infty) \to [0, +\infty]$, if $g_n(\lambda) \uparrow g(\lambda)$ for every $\lambda \ge 0$, then

$$\int_0^{+\infty} g_n(\lambda)\, \mathrm{d}\lambda \uparrow \int_0^{+\infty} g(\lambda)\, \mathrm{d}\lambda.$$

Proof of Proposition 7.8. Monotonicity implies $g_n \le g_{n+1} \le g$, so

$$\int_0^{+\infty} g_n(\lambda)\, \mathrm{d}\lambda \le \int_0^{+\infty} g_{n+1}(\lambda)\, \mathrm{d}\lambda \le \int_0^{+\infty} g(\lambda)\, \mathrm{d}\lambda.$$

Set $A \coloneqq \lim_{n\to\infty} \int_0^{+\infty} g_n(\lambda)\,\mathrm{d}\lambda$. For any fixed finite partition P, the lower sum contains finitely many non-negative terms, so

$$L(g;P) = \lim_{n\to\infty} L(g_n;P) \le \lim_{n\to\infty} \int_0^{+\infty} g_n(\lambda)\,\mathrm{d}\lambda = A.$$

Taking the supremum over all finite partitions P gives $\int_0^{+\infty} g(\lambda)\,\mathrm{d}\lambda \le A$. Combining this with the inequality $A \le \int_0^{+\infty} g(\lambda)\,\mathrm{d}\lambda$ proves the result. □

Proposition 7.9 (Increasing superlevel measures). For measurable functions $f_n \colon X \to [0,+\infty]$, if $f_n(x) \uparrow f(x)$ for every $x \in X$, then for every $\lambda \ge 0$,

$$\mu(\{x \in X \colon f_n(x) > \lambda\}) \uparrow \mu(\{x \in X \colon f(x) > \lambda\}).$$

Proof of Proposition 7.9. Fix $\lambda \ge 0$. If $f(x) > \lambda$, the assumption $f_n(x) \uparrow f(x)$ ensures $f_n(x) > \lambda$ for some n. Therefore,

$$\{x \in X \colon f_n(x) > \lambda\} \uparrow \{x \in X \colon f(x) > \lambda\}.$$

Continuity of the measure μ from below yields

$$\mu(\{x \in X \colon f_n(x) > \lambda\}) \uparrow \mu(\{x \in X \colon f(x) > \lambda\}).$$

□

Theorem 7.10 (Lebesgue monotone convergence theorem). For measurable functions $f_n \colon X \to [0,+\infty]$, if $f_n(x) \uparrow f(x)$ for every $x \in X$, then f is measurable and

$$\int_X f_n\,\mathrm{d}\mu \uparrow \int_X f\,\mathrm{d}\mu. \tag{7.5}$$

Both sides of (7.5) may equal $+\infty$.

Proof of Theorem 7.10. The pointwise supremum $f = \sup_n f_n$ is measurable. For $\lambda \ge 0$, define

$$g_n(\lambda) \coloneqq \mu(\{x \in X \colon f_n(x) > \lambda\}), \qquad g(\lambda) \coloneqq \mu(\{x \in X \colon f(x) > \lambda\}).$$

Each function g_n and g is non-increasing. By Proposition 7.9, $g_n(\lambda) \uparrow g(\lambda)$ for every $\lambda \ge 0$. Applying Proposition 7.8 gives

$$\int_X f_n\,\mathrm{d}\mu = \int_0^{+\infty} g_n(\lambda)\,\mathrm{d}\lambda \uparrow \int_0^{+\infty} g(\lambda)\,\mathrm{d}\lambda = \int_X f\,\mathrm{d}\mu.$$

□

7.4 Positive linearity

To extend positive linearity to general non-negative measurable functions, we approximate each function from below by simple functions and apply monotone convergence to the simple-function identity in Proposition 7.7.

Problem 7.11 (Simple approximation from below). For a measurable function $f\colon X \to [0, +\infty]$, define

$$s_n(x) \coloneqq 2^{-n} \lfloor 2^n \min\{f(x), n\} \rfloor .$$

Prove that every s_n is a non-negative measurable simple function and that $s_n \uparrow f$ pointwise. Prove also that if f is bounded, then $s_n \to f$ uniformly.

Truncation at level n gives s_n finite range, and the floor function rounds down to grid intervals of length 2^{-n}. As n increases, truncation levels grow and grids refine, yielding $s_n \leq s_{n+1} \leq f$. For points where $f(x) < +\infty$, the pointwise error satisfies $f(x) - s_n(x) < 2^{-n}$ for all sufficiently large n. For points where $f(x) = +\infty$, $s_n(x) = n \to +\infty$. When f is bounded, the pointwise error estimate is uniform once $n \geq \sup_X f$.

Theorem 7.12 (Positive linearity of the non-negative integral). For measurable functions $f, g\colon X \to [0, +\infty]$ and constants $a, b \geq 0$,

$$\int_X (af + bg)\, \mathrm{d}\mu = a \int_X f\, \mathrm{d}\mu + b \int_X g\, \mathrm{d}\mu. \tag{7.6}$$

Equality holds in $[0, +\infty]$ under the convention $0 \cdot (+\infty) = 0$.

Proof of Theorem 7.12. Choose sequences of non-negative measurable simple functions $s_n \uparrow f$ and $t_n \uparrow g$ as in Problem 7.11. Each function $as_n + bt_n$ is simple, and $as_n + bt_n \uparrow af + bg$ pointwise. Positive linearity for simple functions (Proposition 7.7) and the monotone convergence theorem (Theorem 7.10) yield

$$\begin{aligned}
\int_X (af + bg)\, \mathrm{d}\mu &= \lim_{n\to\infty} \int_X (as_n + bt_n)\, \mathrm{d}\mu \\
&= \lim_{n\to\infty} \left(a \int_X s_n\, \mathrm{d}\mu + b \int_X t_n\, \mathrm{d}\mu \right) \\
&= a \int_X f\, \mathrm{d}\mu + b \int_X g\, \mathrm{d}\mu.
\end{aligned}$$

□